王天富

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致力于构建具有真实世界影响力的助力人类成长的应用生态，包括学习、效率和创造等场景。

- 研究兴趣主要包括 Agentic AI、大模型、强化学习与数据挖掘，关注相关技术基础与应用。
- 热衷于探索创新应用与可落地算法方案的实践型研究，乐于分析与解决应用场景中实际需求。

一作 CCF-A 论文 6 篇 | 在投一作 4 篇 | 总 A 类论文 10+ 篇 | 总引用量 300+ | Github Stars 600+

教育背景

香港科技大学（广州）HKUST-GZ - 信息学域 - 人工智能专业 2025 – 至今

博士 | 师从熊辉教授（Fellow of AAAI, AAAS, ACM &IEEE）与袁晶博士（IEEE Fellow & 微软合伙人）

中国科学技术大学 USTC - 计算机科学与技术学院 - 计算机科学与技术专业 2022 – 2025

硕士 | 推免至认知智能全国重点实验室，师从熊辉教授（香港科技大学（广州）协理副校长）

重庆大学 CQU - 大数据与软件学院 - 软件工程专业 2018 – 2022

学士 | 专业排名为 6/245（前 3%）；GPA 为 3.78/4.00；加权平均分为 90.11；优异生联盟成员

实习经历

腾讯 CodeBuddy - 青云计划研究实习生（导师：董汉德） 2026.01 – NOW

研究 Coding 场景下自演化的 Multiagent System 算法；构建基于人类反馈的 Agent Team 管理产品

微软 AI - 研究实习生（导师：袁晶博士） 2024.05 – 2025.10

专注于研发面向教育与社会场景的创新型 LLM 智能体系统。 大模型 智能体 强化学习

- WWW'25 | 目标导向的职业知识导师系统，端到端实现目标解析、技能诊断与课程规划的教学。
- KDD'26 审稿中 | 个性化的社交技能学习平台，基于强化学习自适应调度情境，融合游戏化教学。
- 开发了用于提升大模型情商及社会智能的智能体强化学习方法，Agentic RL 提升社交目标完成度。
- 研究应用于数字员工平台的自进化智能体框架，包括 Prompt 优化、Memory 演化和 Agentic RL。

微软 AI & MSRA - 研究实习生（导师：袁晶博士 & 练建勋博士） 2022.08 – 2023.12

构建 Web3 数据基础设施和算法研发，参与 NFT 产品后端开发。 图神经网络 区块链 多模态

- KDD'24 | 提出基于时序图的 NFT 钱包画像算法，建模多行为交易模式，提升价格预测的准确性。
- VLDB'25 | 提出风险可控的多目标投资组合算法，通过风险插值与收益提升机制优化跨市场效益。
- MM'23 | 开发结合视觉策略奖励强化学习的 Prompt 增强算法，实现市场价值感知的 NFT 生成。
- 参与开发了微软内部 Web3 数据基建，爬取链上链下数据，用于 Web3 搜索和 PowerNFT 等产品。

京东探索研究院 - 研究实习生 (导师: 沈力教授) 2021.08 – 2022.04

研究云计算场景中的组合优化机器学习方法。 强化学习 图神经网络 组合优化

- TSC'23 | 提出基于层次强化学习的联合资源分配与调度方法，显著提升系统长期资源利用效率。

获奖情况

安徽省优秀毕业生 （*1%，2025*）；**重庆市本科优秀毕业论文** （*1%，2022*）；华为智能基座未来之星（*2%，2021*）；重庆大学优异生、优秀学生、优秀学生干部、青年先进志愿者（*5%，2022,2019,2020,2020*）

研究生国家奖学金 （*2%，2024*）；**本科生国家奖学金** （*2%，2021*）；朱敬文奖学金 （*2%，2020*）
中科大研究生学术奖学金 *3 次 (2022-2024)*；重庆大学优秀本科生奖学金 *4 次 (2018-2022)*

ICML'25 Workshop on ML4Wireless* 最佳论文奖 *(1/ 33+，2025)；**中国高校计算机大赛网络技术挑战赛全国一等奖 *(3/ 1000+, 2021)***；其他竞赛奖项全国二、三等奖 * 5（如数模美赛 M 奖、蓝桥杯三等奖）

技能爱好

- **算法**: 大模型智能体（LLM Agent）；强化学习（RL）；图神经网络（GNN）；组合优化；多模态模型
- **工程**: 后端（Django,SpringBoot）；前端（Vue,CSS,HTML）；数据库（SQL）；智能合约（Solidity）
- **其他**: 熟悉 PPT 制作、视频剪辑、Figma 设计；英语六级、雅思 6.5；爱好摄影、马拉松、乒乓球

其他情况

- 曾参加剑桥大学人工智能交流项目（2021）和东京大学、早稻田大学智能计算访学项目（2020）
- 独立开发了网络资源分配框架 Virne（Star 100+），含 20+ 强化学习、演化计算、启发式算法
- 知名开源组织 Datawhale 意向成员，《统计学习方法》习题解答项目核心贡献者（Star 1.9K+）

SELECTED PUBLICATIONS [GOOGLE SCHOLAR , DBLP]

[1] **Tianfu Wang**, Yi Zhan, Jianxun Lian, Zhengyu Hu, Nicholas Jing Yuan, Qi Zhang, Xing Xie, and Hui Xiong. LLM-powered Multi-agent Framework for Goal-oriented Learning in Intelligent Tutoring System. In *ACM Web Conference (**WWW**)*, 2025. (CCF-A, CORE A*, Oral Presentation) **Fully deployed by the UK AI4Ed startup *grasp.study* and Received a direct job opportunity invitation from its CEO**.

[2] **Tianfu Wang**, Max Xiong, Shang Qin, Jianxun Lian, Hongyuan Zhu, Zhengyu Hu, Yuxuan Lei, Nicholas Jing Yuan, and Qi Zhang. SocialCoach: Personalized Social Skill Training with LLM-based Agentic Tutoring and Practice. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2026. (CCF-A, CORE A*, Under Review) Deployed in EQoach.

[3] **Tianfu Wang**, Leilei Ding, Ziyang Tao, Yi Zhan, Wei Wu, Yuxuan Lei, Yizhao Xu, Hongyuan Zhu, Qi Liu, Yanyong Zhang, Nicholas Jing Yuan, and Hui Xiong. EvoDiagram: Agentic Editable Diagram Creation via Design Expertise Evolution. In *International Conference on Machine Learning (**ICML**)*, 2026. (CCF-A, CORE A*, Under Review).

[4] **Tianfu Wang**, Xichong Zhang, Yuan Feng, Wei Wu, Zhengyu Hu, Yili Wang, Yijie Xu, Yudong Zhang, and Hui Xiong. From Intent to Agreement: The Agentic Consensus Layer for Collaborative and Auditable Human–AI Coding Engineering. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2026. (CCF-A, CORE A*, Under Review).

[5] **Tianfu Wang**, Liwei Deng, Xi Chen, Junyang Wang, Huiguo He, Leilei Ding, Wei Wu, Qilin Fan, and Hui Xiong. Virne: A Comprehensive Benchmark of RL-based Network Resource Allocation in NFV. In *International Conference on Learning Representations (**ICLR**)*, 2026. (CORE A*).

[6] **Tianfu Wang**, Long Yang, Chao Wang, Chuan Qin, Liwei Deng, Wei Wu, Li Shen, and Hui Xiong. CONAL: Towards Constraint-aware Learning for Resource Allocation in NFV-enabled Networks. In *IEEE Transactions on Mobile Computing (**TMC**)*, 2026. (CCF-A, CORE A*, Under Review) **Best Paper Award in ML4Wireless@ICML25 Workshop** (1 of 33 accepted papers).

[7] **Tianfu Wang**, Liwei Deng, Chao Wang, Jianxun Lian, Yue Yan, Nicholas Jing Yuan, Qi Zhang, and Hui Xiong. COMET: NFT Price Prediction with Wallet Profiling. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2024. (CCF-A, CORE A*) Deployed in Microsoft NFT Product, PowerNFT.

[8] Liwei Deng*, **Tianfu Wang***, Yan Zhao, and Kai Zheng. MILLION: A General Multi-Objective Framework with Controllable Risk for Portfolio Management. In *International Conference on Very Large Data Bases (**VLDB**)*, 2025. (CCF-A, CORE A*, Equal Contribution).

[9] **Tianfu Wang**, Qilin Fan, Chao Wang, Leilei Ding, Nicholas Jing Yuan, and Hui Xiong. FlagVNE: A Flexible and Generalizable Reinforcement Learning Framework for Network Resource Allocation. In *International Joint Conference on Artificial Intelligence (**IJCAI**)*, 2024. (CCF-A, CORE A*).

[10] **Tianfu Wang**, Shen Li, Qilin Fan, Tong Xu, Tongliang Liu, and Hui Xiong. Joint Admission Control and Resource Allocation of Virtual Network Embedding via Hierarchical Deep Reinforcement Learning. In *IEEE Transactions on Services Computing (**TSC**)*, 2023. (CCF-A, CORE A*, JCR-Q1).

SERVICES

- **Reviewer**: ICML'25-26; ICLR'25-26; NeurIPS'24-25; KDD'25-26; WWW'24; AAAI'26; MM'23-24; IJCAI'26; AISTATS'25-26; IEEE TNSM, TNSE, IoTJ

研究总结

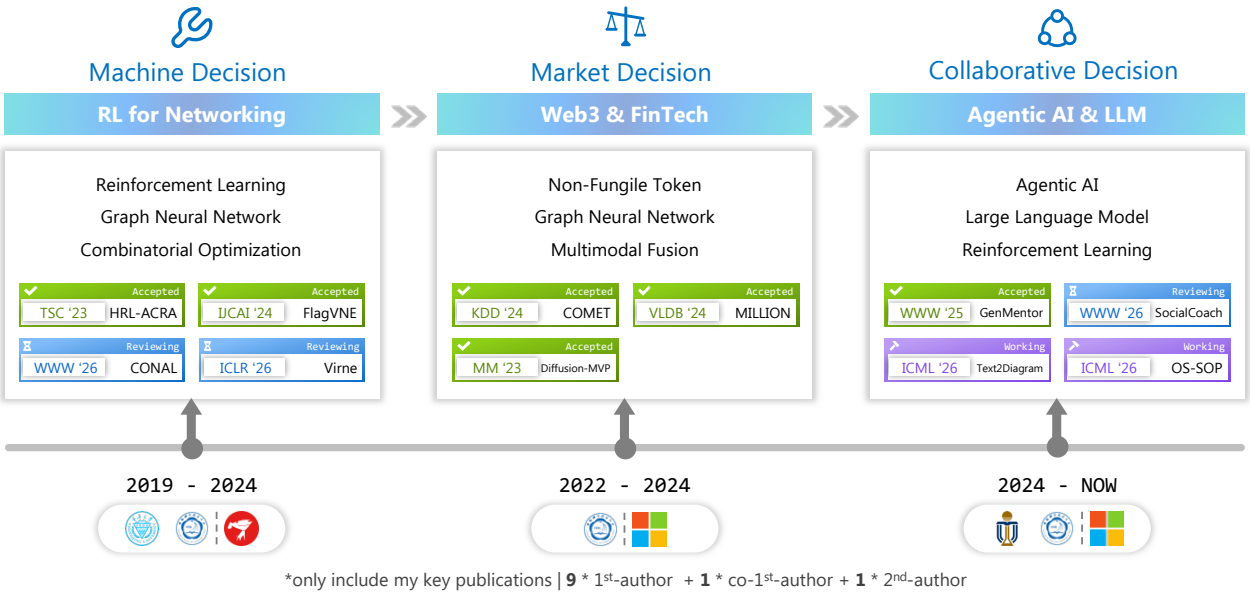

Figure 1: 研究轨迹

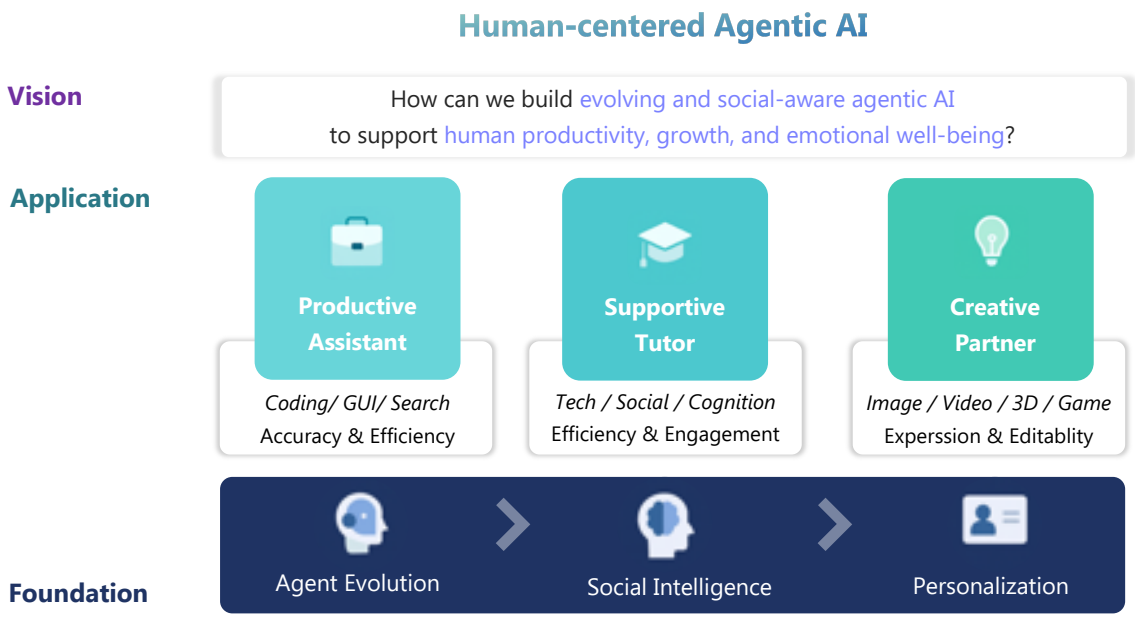

Figure 2: 研究愿景

博士 | Human-centered Agentic AI 基础与应用

@AuraX.ai 人机协作友好的生成式创意设计 Agent Graphics Benchmark

- ICML '26 审稿中 **EvoDiagram**，一个生成可编辑图表的智能体框架。针对像素生成失控与代码合成门槛高的表征鸿沟，提出基于中间画布模式的多智能体协作系统，实现语义逻辑、视觉风格与空间布局的解耦生成；设计层级式设计知识演化机制，自动生成专家级设计先验；同时发布 CanvasBench 基准，实验表明该系统在语义保真度、结构一致性及交互性上显著优于现有基准。

@ 微软 · 中科大 面向智慧教育的智能体导师系统 LLM RL Agent

- WWW '25 Oral **GenMentor**，一个由 LLM 驱动的多智能体职业教育系统。首先，构建推理增强的目标—技能数据集并微调大模型，其次，实现精准目标解析与技能差距识别；同时，基于动态学习者画像设计可进化学习路径优化器；最后，提出探索—草拟—整合的内容生成机制实现个性化材料定制。系统已在微软与 Grasp.study 等产品落地，显著提升目标对齐与学习效率。
- KDD '26 审稿中 **SocialCoach**，一个面向社交技能学习的智能体式系统。首先，多智能体自动化抽取社交技能理论知识，并构建理论到情境的教学语料；其次以强化学习优化场景练习调度，增

强学习者参与度；同时融合沉浸式目标驱动练习、因果式熟练度诊断与反思式辅导。该系统在微软 EQoach 产品中部署，为用户提供游戏化趣味性的社交技能智能辅导，缩小“知行鸿沟”。

@ 微软 · 中科大 智能体强化学习增强大模型情商及社会智能 LLM RL Agent

- **SocialGoalAgent**，一个大模型社会智能评估与增强框架。首先将社会互动建模为 POMDP，环境支持角色设定、目标驱动情境模拟与多轮对话互动。系统性分析不同 LLM 在社会智能差异，揭示模型规模、推理与对手强度对社会智能表现的影响。进一步设计多轮强化学习方法，引入情绪理解推理作辅任务和情境生成反馈，提升大模型在复杂社交情境中的目标达成率与情绪理解能力。

本科 & 硕士 | 网络场景下的组合优化强化学习

@ 中科大 · 京东 用于网络资源分配的强化学习算法 RL GNN

- IJCAI '24 **FlagVNE**，一种灵活且具泛化性的强化学习资源分配框架。具备双向动作结构、分层解码机制与元强化学习课程训练，实现高效探索与跨规模强泛化，显著提升策略搜索能力和泛化性。
- TSC '23 **HRL-ACRA**，一个联合优化准入控制与资源分配的分层强化学习系统。通过平均奖励优化、多目标内在奖励机制及基于时序感知的图神经网络的拓扑编码提升接受率与长期收益。
- WWW '26 审稿中 **CONAL**，一种面向约束的学习框架，融合容错 CMDP、可达性引导的优化机制、自适应可行性预算以及约束感知图感知结构，实现高度稳定且持续零违约的解决方案。

@ 重庆大学 · 中科大 网络资源分配的仿真与基准构建 RL GNN

- ICLR '26 审稿中 **Virne**，一个面向深度强化学习的 NFV 资源分配综合基准及仿真环境。提供可定制模拟（如端边云、5G 场景）、统一的 gym 环境，以及 30+ 算法（包括启发式、进化算法、RL 方法）的标准实现。开展了系统化的实验分析与性能比较，为 RL 网络优化提供关键洞察。

硕士 | Web3 中的金融科技与收益驱动型 NFT 挖掘

@ 微软 · 中科大 收益感知的 NFT 价格预测与生成 GNN MM

- KDD '24 **COMET**，一个基于钱包画像的 NFT 价格预测方面。首先构建多模态 NFT 元数据及历史交易数据集，然后分析钱包状态与交易历史构建时序多行为交易图，进而通过时序图神经网络建模钱包投资者行为、社区影响等，最后统一实现 Collection 级与 Token 级 NFT 价格预测。该系统部署于微软 NFT 产品，PowerNFT，提供钱包画像数据分析和显著提升在线 NFT 价格预测精度。
- MM '23 **Diffusion-MVP**，一个收益感知的 NFT 图像生成框架。首先构建 NFT-1.5M 领域数据集，用于微调 Stable Diffusion 适配 NFT 风格。其次，引入多种视觉策略奖励（稀有度、美学度、相关性），并基于强化学习训练 LLM Prompt 改写器。该算法被部署在微软 NFT 产品，PowerNFT，在视觉质量与市场价值对齐方面表现显著优异 DELLA · 3 等模型。

@ 微软 · 中科大 具备风险可控性的多目标投资组合优化 RL GNN

- VLDB '25 **MILLION**，一个多目标深度学习框架，通过插值与改善机制增强投资组合的泛化能力，并实现理论支撑下的细粒度风险控制。在多类股票市场中均取得优越的收益-风险表现。

FULL LIST OF PUBLICATIONS [GOOGLE SCHOLAR , DBLP]

Ph.D. | **Foundation and Applications of Human-centered Agentic AI** (LLM,RL)

[1] **Tianfu Wang**, Yi Zhan, Jianxun Lian, Zhengyu Hu, Nicholas Jing Yuan, Qi Zhang, Xing Xie, and Hui Xiong. LLM-powered Multi-agent Framework for Goal-oriented Learning in Intelligent Tutoring System. In *ACM Web Conference (**WWW**)*, 2025. (CCF-A, CORE A*, Oral Presentation) **Fully deployed by the UK AI4Ed startup *grasp.study* and Received a direct job opportunity invitation from its CEO**.

[2] **Tianfu Wang**, Max Xiong, Shang Qin, Jianxun Lian, Hongyuan Zhu, Zhengyu Hu, Yuxuan Lei, Nicholas Jing Yuan, and Qi Zhang. SocialCoach: Personalized Social Skill Training with LLM-based Agentic Tutoring and Practice. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2026. (CCF-A, CORE A*, Under Review) Deployed in EQoach.

[3] **Tianfu Wang**, Leilei Ding, Ziyang Tao, Yi Zhan, Wei Wu, Yuxuan Lei, Yizhao Xu, Hongyuan Zhu, Qi Liu, Yanyong Zhang, Nicholas Jing Yuan, and Hui Xiong. EvoDiagram: Agentic Editable Diagram Creation via Design Expertise Evolution. In *International Conference on Machine Learning (**ICML**)*, 2026. (CCF-A, CORE A*, Under Review).

[4] **Tianfu Wang**, Xichong Zhang, Yuan Feng, Wei Wu, Zhengyu Hu, Yili Wang, Yijie Xu, Yudong Zhang, and Hui Xiong. From Intent to Agreement: The Agentic Consensus Layer for Collaborative and Auditable Human–AI Coding Engineering. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2026. (CCF-A, CORE A*, Under Review).

[5] **Tianfu Wang**, Xichong Zhang, Yuan Feng, Wei Wu, Zhengyu Hu, Yili Wang, Yijie Xu, Yudong Zhang, and Hui Xiong. Make Benchmark Hard Again: Question Distractor Option Generation via Agentic Reinforcement Learning. In, 2026. (Working).

[6] Yuxuan Lei, **Tianfu Wang**, Jianxun Lian, Zhengyu Hu, Defu Lian, and Xing Xie. HumanLLM: Towards Personalized Understanding and Simulation of Human Nature. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2026. (CCF-A, CORE A*).

[7] Zhengyu Hu, Linxin Song, Jieyu Zhang, Zheyuan Xiao, **Tianfu Wang**, Zhenyu Chen, Jianxun Lian, Nicholas Jing Yuan, Kaize Ding, and Hui Xiong. Unveiling the Learning Mind of Language Models: A Cognitive Framework and Empirical Study. In *Annual Conference on Neural Information Processing Systems (**NeurIPS**)*, 2025. (CCF-A, CORE A*).

[8] Wei Wu, Zhuoshi Pan, Chao Wang, Liyi Chen, Yunchu Bai, **Tianfu Wang**, Kun Fu, Zheng Wang, and Hui Xiong. TokenSelect: Efficient Long-Context Inference and Length Extrapolation for LLMs via Dynamic Token-Level KV Cache Selection. In *Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2025. (CCF-B, CORE A*).

[9] Zhengyu Hu, Linxin Song, Jieyu Zhang, Zheyuan Xiao, **Tianfu Wang**, Zhenyu Chen, Jianxun Lian, Nicholas Jing Yuan, Kaize Ding, and Hui Xiong. Explaining Length Bias in LLM-Based Preference Evaluations. In *Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2025. (CCF-B, CORE A*).

[10] Yi Zhan, Qi Liu, Weibo Gao, Zheng Zhang, **Tianfu Wang**, Zhenya Huang, Junyu Lu, and Shuanghong Shen. CoderAgent: Simulating Student Behavior for Personalized Programming Education with Large Language Models. In *International Joint Conference on Artificial Intelligence (**IJCAI**)*, 2025. (CCF-A, CORE A*).

[11] Dapeng Hu, Zhongyang Li, Hongyuan Zhu, **Tianfu Wang**, Nicholas Jing Yuan, and Qi Zhang. Agentic Evolution: A Survey of Human–AI Co-Evolving Systems. In *Technical Report (**Microsoft**)*, . (Working).

[12] Zhengyu Hu, Zheyuan Xiao, Max Xiong, Yuxuan Lei, **Tianfu Wang**, Jianxun Lian, Kaize Ding, Ziang Xiao, Nicholas Jing Yuan, and Xing Xie. Population-Aligned Persona Generation for LLM-based Social Simulation. In *Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2026. (CCF-A, CORE A*, Under Review).

[13] Junyang Wang, Lan Zhang, Zhaomeng Zhou, Mu Yuan, **Tianfu Wang**, and Puhan Luo. AgentRouter: Think Whole or Think Apart? Memory-Augmented Adaptive Routing for Cost-Effective On-Device Agent Inference. In *Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2026. (CCF-A, CORE A*, Under Review).

[14] Wei Wu, Liyi Chen, Congxi Xiao, **Tianfu Wang**, Qimeng Wang, Chengqiang Lu, Yan Gao, YIWU, Yao Hu, and Hui Xiong. Anti-Length Shift: Dynamic Outlier Truncation for Training Efficient Reasoning Models. In *Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2026. (CCF-A, CORE A*, Under Review).

[15] Leilei Ding, Shaoxuan Wang, Dazhong Shen, Ziyang Tao, **Tianfu Wang**, Wuyang Zhang, and Yanyong Zhang. PreemptGuard: General Inference Scheduling for Efficient LLM Inference with Short-Tail Length Prediction. In *ACM Web Conference (**WWW**)*, 2026. (CCF-A, CORE A*, Under Review).

[16] Yin Wu, **Tianfu Wang**, Zechao Chen, Zhengxuan Zhang, Yuyu Luo, Hui Xiong, and Nan Tang. Why Train When You Can Evolve?: Multi-Objective LLM Adaptation Without Gradients. In *International Conference on Machine Learning (**ICML**)*, 2026. (CCF-A, CORE A*, Under Review).

[17] Yuxuan Lei, Jianxun Lian, Defu Lian, Jincenzi Wu, **Tianfu Wang**, and Xing Xie. A Computational Framework for Evaluating Human-likeness in LLMs' Situated Behaviors. In *International Conference on Machine Learning (**ICML**)*, 2026. (CCF-A, CORE A*, Under Review).

[18] Zhangyi Hu, Chenhui Liu, Tian Huang, **Tianfu Wang**, and Yutao Yue. CoSPlay: Cooperative Training-Free Self-Play with Purely Self-Generated Idea-Level Unit Test Attack and Iterative Code Refinement. In *International Conference on Machine Learning (**ICML**)*, 2026. (CCF-A, CORE A*, Under Review).

[19] Huizai Yao, Yijie Xu, **Tianfu Wang**, Yili Wang, Yi Cui, Yunfan Lu, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun, and Hui Xiong. Gated Test-Time Scaling for GUI Grounding via Multi-Seed Zoom Search. In *International Conference on Machine Learning (**ICML**)*, 2026. (CCF-A, CORE A*, Under Review).

B.E. & M.S. | **RL for Combinatorial Optimization on Networking** (RL,GNN)

[1] **Tianfu Wang**, Qilin Fan, Chao Wang, Leilei Ding, Nicholas Jing Yuan, and Hui Xiong. FlagVNE: A Flexible and Generalizable Reinforcement Learning Framework for Network Resource Allocation. In *International Joint Conference on Artificial Intelligence (**IJCAI**)*, 2024. (CCF-A, CORE A*).

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M.S. | **FinTech and Profit-driven NFT Mining in Web3** (RL,GNN)

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Other Topics (*e.g.*, RecSys, VectorDB, 3D Gen, FedLearning)

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